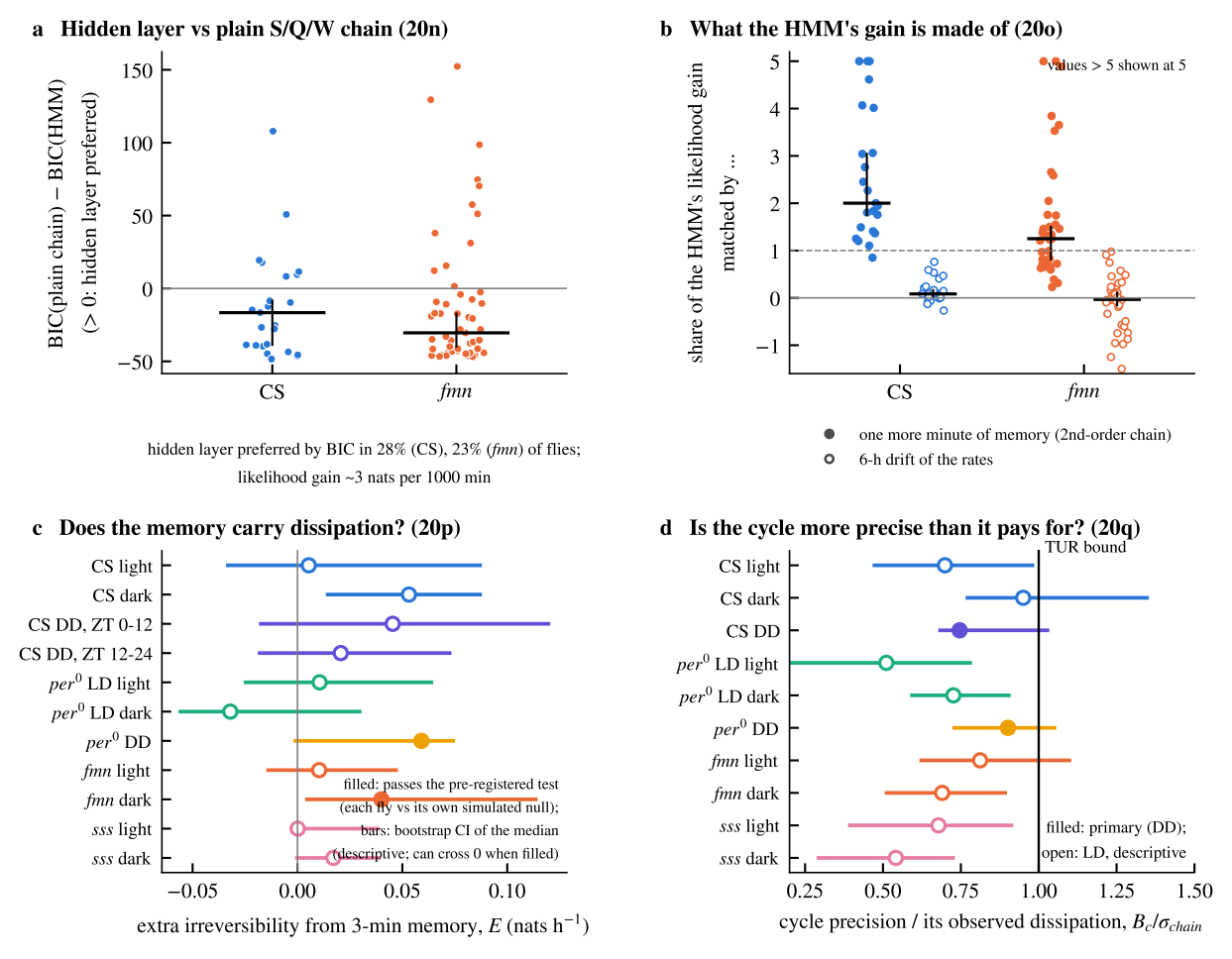

a Hidden layer vs plain S/Q/W chain (20n)
BIC(plain chain) − BIC(HMM)
(> 0: hidden layer preferred)
150
100
50
0
−50
CS
fmn
hidden layer preferred by BIC in 28% (CS), 23% (fmn) of flies;
likelihood gain ~3 nats per 1000 min
b What the HMM's gain is made of (20o)
share of the HMM's likelihood gain
matched by ...
values > 5 shown at 5
5
4
3
2
1
0
−1
CS
fmn
one more minute of memory (2nd-order chain)
6-h drift of the rates
c Does the memory carry dissipation? (20p)
CS light
CS dark
CS DD, ZT 0-12
CS DD, ZT 12-24
per0 LD light
per0 LD dark
per0 DD
fmn light
fmn dark
sss light
sss dark
filled: passes the pre-registered test
(each fly vs its own simulated null);
bars: bootstrap CI of the median
(descriptive; can cross 0 when filled)
−0.05
0.00
0.05
0.10
extra irreversibility from 3-min memory, E (nats h−1)
d Is the cycle more precise than it pays for? (20q)
TUR bound
CS light
CS dark
CS DD
per0 LD light
per0 LD dark
per0 DD
fmn light
fmn dark
sss light
sss dark
filled: primary (DD);
open: LD, descriptive
0.25
0.50
0.75
1.00
1.25
1.50
cycle precision / its observed dissipation, Bc/σchain